Shuhao LI

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INTRO

I am an MPhil student in Civil Engineering at The Hong Kong University of Science and Technology (HKUST). I have been selected as a **University Nominee for the Hong Kong PhD Fellowship Scheme (HKPFS)** and am a recipient of the **HKUST RedBird PhD Recruitment Award**. My research focuses on the application of Artificial Intelligence (AI) in smart buildings, specifically exploring Model Predictive Control (MPC), Deep Reinforcement Learning (DRL), and Large Language Models (LLMs) to optimize building energy performance and operational automation.

My work aims to bridge the gap between advanced computational algorithms and real-world engineering deployment. To this end, I have developed a data-driven MPC framework that achieved **11.52% energy savings** in a Shanghai commercial complex and created **AutoLFM**, a multi-agent LLM framework for automated building load forecasting. Recently, I led my team to win the **Bronze Award (Sustainability & ESG Track)** at the **Hong Kong Techathon+ 2026**, securing **HK$100,000** in incubation funding.

Prior to HKUST, I graduated with honors from Southeast University (ranked 3rd/29), where I was awarded the **National Scholarship (Top 0.2%)**. I have established a strong track record of academic excellence, having published **5 first-author papers** in leading journals such as *Energy and Buildings* and *Building Simulation* . Beyond research, I have demonstrated exceptional leadership as a former President of a student science association and by winning several **National First Prizes** in prestigious competitions, including the National College Student Structural Design Competition.

EDUCATION

2024.09-Present **Hong Kong University of Science and Technology** ***M.Phil. in Civil Engineering***

- CGA 3.67/4.30
- **Selected Honors:** University Nominee for **Hong Kong PhD Fellowship Scheme (HKPFS)** 2026-27

2020.09-2024.07 **Southeast University** ***Bachelor in Intelligent Building***

- GPA 3.85/4.00
- Recipient of the *National Scholarship* (2022)

AWARD

Bronze Award, Hong Kong Techathon+ 10th Anniversary Edition *Jan 2026*

- *Track:* Sustainability & ESG.
- Awarded **HK$100,000** seed funding for project incubation and startup development.

HKUST RedBird PhD Recruitment Award *Jan 2026*

- Recognized for academic excellence and leadership skills.
- Awarded HK$40,000 and a full tuition fee waiver.

National Scholarship (Top 0.2%) *Oct 2022*

Excellent student cadre (Top 1%) *Sep 2022*

Outstanding student title (Top 2%, three times) *Sep 2022*

Jiangsu Province Construction Scholarship *Nov 2021*

LEADERSHIP & ACTIVITIES

President, the Science and Technology Association of the School of Civil Engineering *Jun 2022 —Jun 2023*

- Organized an inter-provincial competition with over 4,000 participants.
- Led a school-level organization with more than 100 members.

- Organized the international competition selection of ASCE Southeast University Chapter.
- Hosted over 30 visits to the Southeast University earthquake platform by professionals from all walks o f life.

PATENT

1. Wang, Z., **Li, S.H.**, Li, S., Chiller Sequencing Optimization Based on Data-Driven Model Predictive Control with Dynamic Programming Method, 202511426158.X
2. Wang, Z., Li, S., **Li, S.H.**, Optimization method, system and product for chiller sequencing, 202410909230.3

PUBLICATION

First Author

1. **Li, S.**, Li, S., Mohebi, P., Wang, D., Ma, M. N., Liu, G., & Wang, Z. (2025). Field demonstration of model predictive control for chiller sequencing in large-scale commercial buildings. Energy and Buildings, 116021. [**IF=7.3**]
2. **Li, S.**, Su, S., & Lin, X. (2025). Optimizing the hyper-parameters of deep reinforcement learning for building control. *Building Simulation* (Vol. 18, No. 4, pp. 765-789). Tsinghua University Press. [**IF=5.9**]
3. **Li, S.**, Li, S., & Wang, Z. (2024). Accelerating chiller sequencing using dynamic programming. *Energy and Buildings*, 325, 115037. [**IF=7.3**]
4. **Li, S.**, Zheng, W., Li, M., Xing, S., & Wang, Z. (2025). AutoLFM: A multi-agent LLM framework for automated building load forecasting. *Building Simulation*. [**IF=5.9**]
5. **Li, S.**, Li, M., Hu, Z., & Wang, Z. (2025). Applying semantic model for easy and fast deployment of chiller sequencing algorithm. *Energy and Buildings*, 116830. [**IF=7.3**]

Co-Author

1. Mohebi, P., **Li, S.**, & Wang, Z. (2025). Chance-constrained stochastic framework for building thermal control under forecast uncertainties. *Energy and Buildings*, *331*, 115385.
2. Su, S., **Li, S.**, Ding, Y., Mao, P., & Chong, D. (2024). Health damage assessment of commuters and staff in the metro system based on field monitoring—A case study of Nanjing. *Frontiers in Public Health*, 11, 1305829.
3. Zheng, W., Liu, L., **Li, S.**, Zhang, Z., Zhao, P., Li, T., ... & Wang, Z. (2026). Smart predict-then-optimize-based model predictive control for Virtual Power Plants with battery storage. *Journal of Energy Storage*, *142*, 119606.
4. Li, M., Hu, Z., Mohebi, P., **Li, S.**, & Wang, Z. (2026). Enhancing LLM-based building data query with chain-of-thought, retrieval-augmented generation, and fine-tuning. *Automation in Construction*, *182*, 106738.

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